Midterm Project:
BWC UX design brief & background research

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Pre-stage research

Before I identify some of the current issues with the BWC software are, I want to first understand these questions:

1. What is BWC?
2. How is BWC used?
3. What role do BWCs play in public safety?
4. What is some current software that BWC use?
    a. How do they work?
    b. What do they look like?
5. What are some current issues or complaints with the BWC software?

What is BWC

Body-worn cameras (BWCs) are small video and audio recorders that police officers and other safety professionals wear on their uniforms. The U.S. National Conference of State Legislatures describes BWCs as "recording devices that police officers wear as part of their uniforms to document what they see during their duties"[1]. They capture interactions with the public and generate digital evidence for investigations, prosecution and accountability. National and state policies emphasize that BWCs should accurately document contacts and help reduce complaints and violent confrontations[2]. However, research shows that the benefits are mixed: cameras can improve transparency and evidence quality, yet they introduce privacy concerns, costs and technical challenges. Modern BWCs are part of integrated systems: devices automatically record and tag footage, software uploads and stores it in secure servers or cloud platforms, and digital evidence management tools allow officials to search, redact and share videos with prosecutors or the public.

How BWCs are used

Modern body-worn cameras are designed to fit into policing workflows from activation through evidence sharing:

1. **Activation and recording** – Officers wear BWCs on their chest, shoulder or glasses. Agencies set policies for when cameras must be activated (e.g., during traffic stops or use-of-force incidents). Some systems rely on the officer pressing a button; others use *automatic triggers* such as vehicle lights and sirens, foot pursuit detection, holster sensors or geofences. i-PRO's BWC4000 can start recording when a light bar or siren is activated[8], while Utility's EOS camera includes a two-minute pre-event buffer and automatic triggers for holster removal, running, or entering an action zone[6].

2. **Tagging and metadata** – During or after recording, officers add metadata (case number, incident type, etc.) via buttons, LCD menus or mobile apps. Motorola's VideoManager allows in-field tagging through its mobile SmartControl app[9].



Figure 1Motorola Solutions, showing AI redaction engine. Can not find images of mobile SmartControl app.

3. **Uploading and off-loading** – After a shift, officers dock their BWCs or use wireless upload features. Utility's EOS camera automatically offloads encrypted evidence via Wi-Fi or cellular networks, eliminating the need to dock[7].

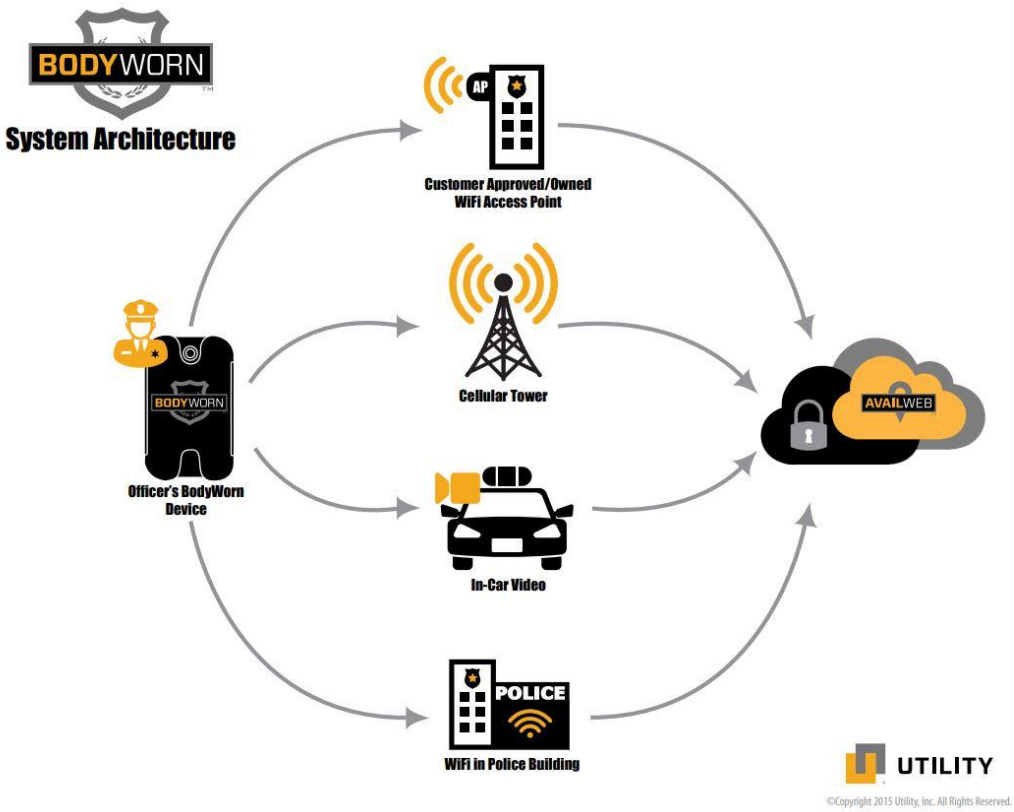


Figure 2 EOS System Architecture

4. **Storage and management** – Evidence is stored on secure servers or cloud platforms. Software platforms like i-PRO's Unified Digital Evidence (UDE) encrypt and hash files, maintain tamper-evident logs and track every open, share and save[10].

5. **Review, redaction and case building** – Investigators review footage, add notes and attach documents. AI-powered redaction tools automatically blur faces and license plates and mute audio. For example, VideoManager EL provides AI-powered redaction and transcription (PC software showcase in Figure 1) to quickly identify and redact sensitive information[9].

6. **Sharing and disclosure** – Evidence is shared internally or externally. Many platforms offer web portals or password-protected links for prosecutors and defense attorneys.

Role of BWCs in Public Safety

As mentioned in the previous sections, BWCs are changing how evidence is recorded/presented, as well as the evidence quality. But it's more than that:

Benefits and evidence

- **Improved documentation and accountability** – BWCs capture objective evidence of interactions, which can deter misconduct and protect officers from false accusations. The Texas model policy notes that cameras provide accurate documentation and may reduce violent confrontations and citizen complaints[1].
- **Enhanced evidence quality** – BWCs record statements and behaviors at the scene. The NIJ reports that agencies adopt BWCs to improve officer safety and evidence collection and to reduce complaints and liability[19].
- **Potential reduction in complaints** – A Campbell systematic review found that BWCs can reduce citizen complaints but that the reduction may reflect either improved interactions or reduced reporting[20]. Some research suggests cameras may lead to fewer unsubstantiated complaints and improved police professionalism.
- **Training and feedback** – Footage can be used for officer training, early intervention and policy evaluation. Agencies like i-PRO's UDE allow playback on mobile devices for in-field review on web-based evidence management[10].

Mixed evidence and controversies

- **Limited effect on use of force** – Research has not found consistent reductions in officer use of force or assaults. The NIJ notes that studies show no statistically significant effects on officer or citizen behaviors[22]. The Campbell review similarly concludes that BWCs have "no clear or consistent effects" on most outcomes and that restricting officer discretion may reduce use of force[23].
- **Cost and resource burdens** – Beyond purchasing cameras, agencies must pay for storage, data management and redaction. A cost-benefit study reports that Dallas Police Department consumed 98 terabytes of storage and that redaction can require 15–30 minutes of staff time per minute of video. Ten minutes of footage may take an hour to redact[24].
- **Privacy concerns** – Civil liberties groups warn that cameras can become surveillance tools. The ACLU argues that without strong policies, departments may use cameras to monitor the public and withhold videos of misconduct[26]. Facial-recognition and real-time analytics raise further concerns about bias and civil rights; the R Street Institute urges strict regulation and human oversight of AI-powered BWCs[28].

Current BWC software

While trying to find some current BWC software, I noticed that most of them are actually Digital Evidence Management System (DEMS) rather than the actual software used on the BWC device. From the design brief, we might need to focus on the software/application of the BWC device. However the actual software/application used for the BWC device depends highly on the brand of the device and can be very different depends on the device

type (such as whether the device has a screen, or what types of features one device provides.) Therefore, in the following section, I will be focus on different brands/models of BWC devices and what features they offer.

Previously, to understand the BWC device and how it's used, I've mentioned these brands/models:

1. VideoManager EL by Motorola
2. i-PRO's BWC4000
3. EOS by Utility – Device (Utility has a very good demonstration video to show how their device work *video link*)
4. Axon Body 4
    a. This one was not mentioned in previous sections, but Axon was commonly seen been used in police report videos.

To learn more about what some of the current BWC devices are, I want to do more research and answer these questions:

1. What are some different types of BWC devices?
2. What are common features of modern BWC devices?
3. What are unique features from specific BWC brands/models?
4. What are common issues or complaints about current BWC devices?
5. What can be improved (features or on-device software) to address those issues?

What are some different types of BWC devices

1. **Chest-mounted / Lapel-mounted**
    a. Cameras attach to the chest or shirt front using magnetic or clip mounts. The placement offers stability and is the most common because it's affordable and easy to attach[29]. However, the view can be partially blocked when officers raise their arms[30].



2. **Shoulder-mounted**
    a. Devices clip onto epaulets or shoulder straps, providing a higher perspective and less obstruction from hands. Shoulder-mounted systems can integrate with uniform radio loops or include external sensors.



3. **Head- or eyewear-mounted**
    a. Cameras are attached to helmets or glasses to capture the officer's line of sight[29]. They provide the most accurate perspective but may be uncomfortable or present a snag hazard[30].



4. **Integrated uniform / badge cameras**
    a. Cameras are built into the uniform, badge or radio so they're less noticeable and harder to knock off[29]. Integration helps prevent tampering but may limit flexibility and may raise privacy concerns.



5. **Smartphone-based solutions**
    a. Rugged smartphones with BWC apps function as cameras and communication devices. Officers can start recording via a dedicated side button, stream live video and upload directly to cloud storage[5].



Common features of modern BWC devices

- **High-resolution video & wide field-of-view -** Most cameras record in 1080 p with wide-angle lenses.
- **Pre-event or buffered recording -** Cameras continuously record into a temporary buffer so that footage from before an officer presses record is saved. The NIJ notes that pre-event buffers typically store 15 seconds to 2 minutes of video[30].
- **Battery life & power management -** Cameras must last a full shift. NIJ recommends at least 3 hours of continuous recording and ideally 8–12 hours[30].
- **Durability -** BWCs are built to withstand harsh conditions. Devices often meet IP67 or IP68 water/dust ratings and military-standard drop tests.
- **Low-light and HDR recording -** NIJ explains that low-light capability may involve infrared or low-lux sensors[30].
- **Audio recording and noise reduction -** Modern cameras capture clear audio. i-PRO BWC4000 uses four microphones with noise reduction[8]. Motorola VB400 records audio throughout its continuous buffer[31].
- **GPS / location & connectivity -** Many devices embed GPS coordinates or support Wi-Fi and Bluetooth. i-PRO BWC4000 features built-in GPS/WLAN/Bluetooth LE[8]; Motorola VB400 pairs with radios via Bluetooth for automatic triggering[31].

- **Event tagging and metadata -** Officers can add classification tags in the field. i-PRO BWC4000 includes an LCD menu for tagging[8].
- **Automatic triggers & sensors -** To reduce reliance on manual activation, cameras integrate with vehicle sensors or external triggers. Axon and Motorola use wireless signals to start recording when a cruiser's light bar is activated[32]; i-PRO BWC4000 can be triggered by a light bar, siren or G-force sensor[8]; Utility EOS uses AI to trigger recording during gunshots, when the officer is prone or enters a geofenced zone[32].
- **Encryption and security -** FIPS 140-2 certified encryption protects data on the device. i-PRO BWC4000 meets FIPS 140-2[8], while Axon Body 4 offers full-disk AES-256 encryption[33].
- **On-device user interface -** Simple controls allow officers to operate with gloves. Motorola VB400 has a large central record button and side switches[31]. Reveal D5 features a front-facing screen to show when recording[32]. Getac BC-03 includes haptic feedback and a rotating category wheel (supports in-field tagging by using the physical wheel)[34].
- **Real-time streaming -** Some devices support live streaming to a command center, enabling supervisors to view events as they unfold. Axon Body 4 uses the Respond service for live video[33]; Motorola VB400 streams via Wi-Fi/Bluetooth[31].

Unique brand/model features

1. **Axon Body 4 (no screen)** [33]
    a. **Real-Time Translation -** supports real-time conversation in more than 50 languages, helping officers communicate clearly and quickly make informed decisions when a human translator isn't available.
    b. **Programmable buttons and feedback -** programmable multi-function button, muted-audio mode with reminders, and status lights simplify operation.
    c. **Report generation -** Draft One (Axon's report-writing system) force multiplies agencies and saves officers multiple hours every week by automatically drafting police report narratives based on body-worn camera audio.
    d. **Flex POV accessory -** optional head- or shoulder-mounted camera attaches to the Body 4, enabling an unobstructed view of the officer's line of sight.
2. **Motorola VB400 (Motorola Solutions)** [31]
    a. **Pre/post-record looping -** continuously records a loop of audio and video, capturing events before and after the record button is pressed.

    b. **Peer-assisted recording -** automatically triggers nearby colleagues' cameras when one camera begins recording, improving coverage in multi-officer events.
    c. **Radio integration -** links with TETRA radios, enabling automatic activation when an emergency button is pressed or a man-down sensor is triggered.
    d. **Mobile application linking -** The VB400 body-worn camera seamlessly integrates with Motorola Solutions Digital Policing Application, Pronto.
3. **i-PRO BWC4000 (Panasonic / i-PRO)** [8]
    a. **LCD tagging interface and 3-minute pre-record:** officers can tag footage on-device and configure up to a three-minute pre-record buffer. (However, the LCD screen is very small on the device, couldn't find demonstration videos therefore not sure how the tagging process could be done.)
4. **EOS by Utility(Utility Associates)** [6]
    a. **Sensors loaded –** EOS has many sensors that allows the device to auto activate, such as holster senser, vehicle sensor, foot pursuit sensor, body position sensor (officer down).

Common issues & complaints about BWC Devices (software related)

1. **Difficulty uploading or syncing footage:** Departments report problems docking cameras and uploading video; the Illinois report notes difficulty uploading footage and cameras failing to power on/off[35]. These issues delay evidence transfer and increase workload.
2. **User interface and accidental activation or deactivation:** Small buttons and complex controls may lead to accidental activation or missed recordings. Officers have reported cameras turning off unexpectedly or restarting; the Illinois report lists random restarts and hardware failures[35].
3. **Privacy and constant recording:** Continuous recording may capture sensitive conversations or personal information. The ACLU warns that body cameras risk becoming surveillance tools if agencies do not implement strict policies on activation and access[26]. AI-triggered cameras like Utility's EOS also raise privacy and bias concerns[28].
4. **Administrative workload for tagging and metadata:** While some devices support tagging, many require officers to manually assign categories, which can be time-consuming. Inadequate on-device interfaces can result in incomplete metadata and hinder search later (inferred from NIJ concerns about training and resources[19]).

Problem Statement

From previous secondary research and design precedent examples I finally understand more about BWC in public safety and can identify the problems that I would like to target. Although the issue I would like to target wasn't mentioned frequently, I think it can improve the one of the most important function of BWC.

Statement

Law-enforcement agencies across the world have started using body-worn cameras (BWCs) to increase transparency, improve evidence quality and protect both officers and civilians. However, the use of BWCs has created a data-management crisis. Agencies record everything from traffic stops to active-shooter responses, and the resulting video volume has ballooned from a few terabytes to petabytes of footage in less than a decade[36]. With the amount of recordings in the system, it becomes hard to find the recordings needed. Tagging helps agencies find the right videos fast so they can respond legally and on time. However, that means Officers must download and tag every video as evidentiary or non-evidentiary and categorize the incident type by the end of each shift[37]. Tagging can be a very tiring task and requires concentration as mis-tag may lead to permanent deletion of important evidence. These duties consume time that could be spent on patrol. Some officers report having to sit for hours to review and tag footage before it is ready for disclosure[37].

The desired future state for this project is a streamlined body-camera management workflow where, after each recording, a dispatcher or designated staff member completes a **simple tagging and classification process using voice input and AI voice recognition and classification to auto apply tags or add notes to the video**. The system automatically stores videos in category-specific repositories, applies policy-based retention schedules, and links footage to case identifiers so that evidence managers can quickly retrieve only the relevant videos. This solution will reduce the number of human hours spent reviewing and tagging footage, lower storage costs by deleting non-evidentiary files on schedule, and improve chain-of-custody integrity. Bridging this gap is significant because it frees officers to focus on public safety while ensuring that digital evidence is organized, searchable and retained appropriately.

Potential risk and concerns

While brainstorming for the solution to the targeted issue, I realized that although AI can reduce human workload, it's not error-free at this point of time. Over-reliance on AI-generated outputs may worsen existing problems, such as incorrect tagging. In addition, the use of AI may raise concerns about data privacy and the risk of information leakage.

Additional Design Precedent

After brainstorming for potential solutions to the targeted issue, I started research again to see if there's any existing solution that already uses AI for tagging footages.

In 2024, Wrap Technologies lunched "WrapAI™". This new technology enables AI video redaction, auto transcripts of voice-captured video, AI assisted contextual content search with AI auto-tagging as their hero feature.

"Leveraging state-of-the-art machine-learning algorithms, WrapAI™'s Auto-Tagging feature automatically identifies and labels objects, activities, and individuals within video content, significantly reducing the time and resources required for manual tagging and analysis of content."

But just like what I worried in the "potential risk and concerns" section, WrapAI seems to aim for fully AI powered functionalities. Though it can greatly reduce human labor and significantly improve efficiency in tagging footages and finding relevant evidence. It's still at risk to over-rely on AI generated/identified content.

What we can do to double ensure the evidence is correctly tagged is to compare the human voice-input with AI generated tags. After an incident, an officer can sit in their vehicle and voice-input what just happened, what was his role, was he involved in shooting, etc.. and the system will auto generate tags according to his voice input. Then, after shift, the officer will upload their footages of the day, and AI systems like Wrap AI can start recognizing the footage and generate/label tags. Then the supervisor system can compare voice-input tags with AI generated tags. If there are no conflicting tags, we have achieved double insurance to ensure tags were labeled correctly. If there are conflicts, then a human review will be required.